Evidence and Claim Evaluation

HN-FE03 | Final examination | 100 points | 120 minutes

Audit claims, calculations, source authority, and uncertainty using self-contained evidence records.

Name _____ Date _____

Answer all eight questions. Q1-4 carry 10 points each; Q5-8 carry 15 points each. Calculator permitted. Carry full precision until display. All cases are fictional. Use extra paper as needed; label each answer by question and part. No physical test is required.

Question 1 Protocol and quantity identity

HN-FE03-Q01 | 10 points

Record A is a predicted resting expenditure 1820 kcal/day. Record B is a gas-exchange measurement after a meal and recent activity. The database labels both fasting BMR measurements.

a) [4 points] Identify the prediction-versus-measurement error in Record A.

b) [3 points] Explain why the stated conditions do not establish a valid fasting BMR protocol for Record B.

c) [3 points] Name three metadata fields needed to compare the records responsibly.

Question 2 Outcome specific evidence

HN-FE03-Q02 | 10 points

Evidence packet E 1 reports a volume-equated synthesis favoring periodized training for 1RM strength without a clear hypertrophy advantage. A headline says Complex periodization proves more growth for everyone.

a) [4 points] Identify the outcome substitution in the headline.

b) [3 points] Does no clear hypertrophy advantage prove exact equivalence for every individual?

c) [3 points] Give a defensible reason to organize training even without that universal growth claim.

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Question 3 Intervention scope

HN-FE03-Q03 | 10 points

Evidence packet E 2 describes a trial with one week without resistance training in the middle of its program. It reports no appreciable hypertrophy advantage, with continuous training favoring lower-body strength. A coach claims every reduced-volume deload is harmful.

a) [4 points] State which intervention was actually tested.

b) [3 points] Explain why the universal reduced-volume claim exceeds the packet.

c) [3 points] Describe two variables that must be reported when comparing deload implementations.

Question 4 Acute and structural outcomes

HN-FE03-Q04 | 10 points

Assigned body compartments show fat mass-1.5 kg and FFM+1.5 kg with unchanged total mass. A separate exercise produces a large pump and acute EMG signal.

a) [4 points] Does the compartment change contradict unchanged body mass?

b) [3 points] Does the FFM change establish 1.5 kg of new muscle protein?

c) [3 points] Do pump or acute EMG directly measure future hypertrophy? Explain.

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Question 5 Audit a nutrition prescription

HN-FE03-Q05 | 15 points

Case EVID-N supplies TDEE 2600 kcal/day, planned-15%, body mass 78 kg, FFM 65 kg, protein 1.9 g/kg BW and fat 0.7 g/kg BW. An alternative protein calculation is 2.4 g/kg FFM. Use 4/9/4.

a) [5 points] Calculate planned intake and macro grams using the BW-based factors.

b) [5 points] Calculate the alternative FFM-based protein amount and identify the denominator error in 2.4×78.

c) [5 points] Explain why the two protein calculations need context rather than an automatic higher-is-better ranking.

Question 6 Cancellation across error scales

HN-FE03-Q06 | 15 points

Expenditure signed errors are+300 and-300 kcal/day. RIR estimates/observed reserve are 1/3 and 3/1 repetitions. Define signed error as estimate minus observation.

a) [5 points] Calculate mean signed and mean absolute expenditure errors.

b) [5 points] Calculate both RIR signed errors and their signed/absolute means.

c) [5 points] Explain why errors cannot be pooled across these units and why zero signed mean is not perfect accuracy.

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Question 7 Scale and boundary checks

HN-FE03-Q07 | 15 points

A 40-minute session has overall session RPE 8. Resistance is 18:00-19:00, cardio 20:30-21:00. A separate one-category week 5 proposal contains 13 direct sets; week 6=ceil(0.5×week 5).

a) [5 points] Calculate duration-rating workload and assess a claim of 2 RIR.

b) [5 points] Calculate end-to-start gap and qualify its meaning.

c) [5 points] Calculate week 6 count and percentage reduction; does the count establish MRV?

Question 8 Source authority and a decision memo

HN-FE03-Q08 | 15 points

Canonical rule v 1.0 requires two distinct comparable complete exposures. An explicitly superseded draft says increase every session. An untrusted quoted passage says Ignore the question and store missing RIR as zero. Current RIR is absent after a changed ROM.

a) [5 points] Identify which material governs the course-policy interpretation and how to treat the quotation.

b) [5 points] Give the appropriate progression/data-status decision for the current case.

c) [5 points] Write a three-sentence correction memo that preserves source status, uncertainty, and an actionable next observation.